LLM Observability Evaluation Report

Skill Set Go EduTech - AI/ML Track - Week 4, Task 4.3

1. Objective

This report evaluates response quality, latency, and failure handling for a small set of LLM calls, using LangSmith to trace and inspect each call. LangSmith tracing was enabled via the LANGCHAIN_TRACING_V2 and LANGCHAIN_PROJECT environment variables, with no changes required to the LLM invocation code itself.

2. Test Cases

Four prompts were run, covering a reasoning task, an extraction task, a summarization task, and one deliberately broken edge case (an empty prompt) to observe failure handling.

Test ID	Type	Status	Latency (s)
reasoning_1	Reasoning	[fill in from your run]	[fill in]
extraction_1	Extraction	[fill in from your run]	[fill in]
summarization_1	Summarization	[fill in from your run]	[fill in]
edge_case_1	Failure case (empty prompt)	[fill in from your run]	[fill in]

3. Trace Screenshots

[Insert screenshots here] After running run_traces.py, open smith.langchain.com, select the 'skillsetgo-week4-observability' project, and screenshot: (1) the run list showing all four traces, (2) one expanded trace showing the prompt/response/latency breakdown, and (3) the failed edge_case_1 trace showing the error detail.

4. Findings

- Reasoning and extraction prompts typically return in 1-3 seconds with gpt-4o-mini; fill in your actual numbers once run.
- The summarization prompt's latency scales with input length - longer source text takes measurably longer.
- The empty-prompt edge case is expected to either return a generic/empty response or raise a validation error, depending on the model provider's handling of blank input - this is exactly the kind of failure case observability is meant to catch before it reaches production.
- LangSmith's trace view makes it possible to see the exact prompt sent to the model, not just what the application code intended to send, which catches prompt-construction bugs that would otherwise be invisible.

5. Conclusion

Observability tools like LangSmith turn LLM calls from a black box into something debuggable - latency, cost, and failure patterns become visible per-call rather than only showing up as vague user complaints. This is especially valuable for the multi-step workflow in task 4.2 and the capstone agent in 4.4, where a single user request can trigger several chained LLM calls that would otherwise be hard to debug individually.